Lakshya JEE AIR (2025)

WEEKLY TEST - 30

DURATION : 180 Minutes | **DATE : 25/11/2024** | **M. MARKS : 300**

ANSWER KEY

PHYSICS

1. (2)
2. (3)
3. (4)
4. (3)
5. (4)
6. (3)
7. (4)
8. (1)
9. (2)
10. (3)
11. (3)
12. (3)
13. (1)
14. (3)
15. (4)
16. (3)
17. (3)
18. (1)
19. (4)
20. (3)
21. (494)
22. (12)
23. (2)
24. (6)
25. (12)

CHEMISTRY

26. (2)
27. (1)
28. (3)
29. (3)
30. (2)
31. (3)
32. (1)
33. (2)
34. (2)
35. (3)
36. (4)
37. (3)
38. (3)
39. (4)
40. (2)
41. (2)
42. (4)
43. (3)
44. (3)
45. (3)
46. (2)
47. (9)
48. (4)
49. (6)
50. (4)

MATHEMATICS

51. (2)
52. (3)
53. (4)
54. (2)
55. (2)
56. (1)
57. (4)
58. (2)
59. (3)
60. (3)
61. (2)
62. (3)
63. (4)
64. (2)
65. (2)
66. (1)
67. (1)
68. (3)
69. (2)
70. (3)
71. (9)
72. (9)
73. (91)
74. (6)
75. (2)

SECTION-I (PHYSICS)

1. **(2)**

Magnetic field is given by the formula $\frac{\mu_0 I}{4\pi a}(\sin\theta_1 + \sin\theta_2)$

2. **(3)**

$$m_P v_e = \frac{h}{\lambda}$$

$$\therefore v_e = \frac{h}{\lambda m_P}$$

$$= \frac{6.63\times10^{-34}}{\left(5200\times10^{-10}\right)\times9.1\times10^{-31}\times1800} = 0.77\,\text{m/sec}$$

3. **(4)**

$$K = \frac{p^2}{2\text{m}}$$

In collision, momentum p remains constant,

$$\therefore K \propto \frac{1}{\text{mass}}$$

After collision, mass has double, So, kinetic energy will remain $\frac{K}{2}$. Hence, loss is also $\frac{K}{2}$.

Now, $\frac{K}{2}$ = minimum excitation energy required.

= 10.2 eV + 10.2 eV

$\Rightarrow$ K = 40.8 eV

4. **(3)**

$W_{1\text{-}2} + W_{2\text{-}3} + W_{3\text{-}1} = -300$ ($\Delta U = 0$ in a cyclic process).

$$= \frac{7}{2}\text{nR}(T_2 - T_1) = 2100\text{R}$$

$$\eta = \frac{-300}{2100R} = -\frac{1}{7R}$$

5. **(4)**

Assume upper hemisphere,

$$B = \frac{2}{3}\pi R^3 \times \rho g$$

$$F_2 = \pi R^2 \times \rho g(2R)$$

$$\therefore \quad F_1 = F_2 - B = \frac{4}{3}\pi R^3 \rho g$$

6. **(3)**

$$kx = 3mg\sin\theta$$

$$kx = 3mg\sin\theta$$

$$x = \frac{3mg\sin\theta}{k}$$

7. **(4)**

It means we are getting only 100 MeV of energy by the fission of one uranium nucleus.

Number of nuclei per second

$$= \frac{\text{Energy required per second}}{\text{Energy obtained by one fission}}$$

$$= \frac{16\times10^6}{100\times1.6\times10^{-13}} = 10^{18}$$

8. **(1)**

$$Q = VC = (C + c)\,V_1 \quad V_1 = \frac{VC}{C+c}$$

$$V_n = \left(\frac{C}{C+c}\right)^n V$$

$$Q_n = V_n C = \left(\frac{C}{C+c}\right)^n VC = \left(\frac{C}{C+c}\right)^n \cdot Q$$

9. **(2)**

$$V = \frac{\rho}{6\varepsilon_0}\left(3R^2 - r^2\right)$$

$$V = \frac{\rho 3R^2}{6\varepsilon_0} - \frac{\rho}{6\varepsilon_0}\left[3\left[\left(\frac{R}{2}\right)^2\right] - \left(\frac{R}{2}\right)^2\right]$$

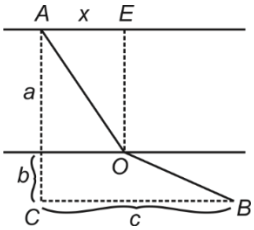
$$= \frac{\rho R^2}{2\varepsilon_0} - \frac{\rho}{6\varepsilon_0}\left[2.\frac{R^2}{4}\right]$$

$$= \frac{\rho R^2}{2\varepsilon_0} - \frac{\rho R^2}{12\varepsilon_0} = \frac{5\rho R^2}{12\varepsilon_0}$$

10. **(3)**

$$T = t_{A\to0} + t_{0\to B}$$

$$T = t_{A\to0} + t_{0\to B}$$



$$T = \frac{\sqrt{a^2 + x^2}}{v_1} + \frac{\sqrt{b^2 + (c-x)^2}}{v_2}$$

for minimum T, $\frac{dT}{dx} = 0$

$$\Rightarrow \quad \frac{\frac{x}{\sqrt{a^2+x^2}}}{\frac{(c-x)}{\sqrt{b^2+(c-x)^2}}} = \frac{v_1}{v_2} \Rightarrow \frac{\sin\alpha_1}{\sin\alpha_2} = \frac{6}{2} = 3$$

11. **(3)**

Two surfaces are formed in the capillary

Since force due to surface tension becomes twice

$\therefore$ $h' = 2h$

12. **(3)**

$\overline{\overline{A}+\overline{B}} = \overline{A.B}$

13. **(1)**

About point *O* ring is in pure rotation.

14. **(3)**

The ray SM after reflection undergoes a phase change of π, for maxima at *P*, path difference between *S* & *S*′ = $\frac{\lambda}{2}$

(*S*′ is virtual source producing back *PM*, symmetric to *S*)

Comparing with YDSE, $d = 4x$, $D = 600x$

Path difference = $\frac{x.d}{D}$

$\frac{\lambda}{2} = \frac{x(4x)}{600\,x} \Rightarrow x = 75\,\lambda$

15. **(4)**

$\overrightarrow{L_i} = \overrightarrow{L_f}$ (About point on horizontal surface)

$\Rightarrow \frac{mR^2}{2}\omega_0 = 2\left(\frac{3}{2}mR^2\right)\omega$

$\Rightarrow \omega = \frac{\omega_0}{6}$

$\Rightarrow V_{cm} = \frac{R\omega_0}{6} \Rightarrow J = \frac{mR\omega_0}{6}$

$\Rightarrow J = \frac{(2)(1)12}{6} = 4\ \text{kg}\cdot\text{m/s}$

16. **(3)**

$\frac{d^2\theta}{dt^2} = -\omega^2\theta$ for angular SHM

$\tau = k\frac{L}{6}\theta\cdot\frac{L}{6} + k\frac{L}{3}\theta\cdot\frac{L}{3}$

$\Rightarrow \left(\frac{2mL^2}{3} + mL^2\right)\cdot\frac{d^2\theta}{dt^2} = -kL^2\theta\left(\frac{1}{36}+\frac{1}{9}\right)$

$\Rightarrow \frac{5mL^2}{3}\cdot\frac{d^2\theta}{dt^2} = -\frac{5kL^2}{36}\cdot\theta$

$\Rightarrow \frac{d^2\theta}{dt^2} = -\frac{k}{12m}\cdot\theta$

$\Rightarrow \omega = \sqrt{\frac{k}{12m}}$

$f = \frac{1}{2\pi}\sqrt{\frac{k}{12m}}$

$= \frac{1}{4\pi}\sqrt{\frac{k}{3m}}$

17. **(3)**

Use dipole

$dq = \lambda_0 \sin\phi\ a\ d\phi$

$|\overrightarrow{dp}| = dq \times 2a - 2a^2\lambda_0 \sin\phi d\phi$

$|\overrightarrow{dp}| = 2a^2\lambda_0 \sin\phi d\phi\left[\cos\phi\hat{i} + \sin\phi\,\hat{j}\right]$

$d\vec{\tau} = \overrightarrow{dp}\times\vec{E}$

$d\vec{\tau} = 2a^2\lambda_0 d\phi\left[\sin\phi\cos\phi\hat{i} + \sin^2\phi\hat{j}\right]\times$

$\left[E_0\hat{i} + E_0\hat{j}\right]$

$\int d\vec{\tau} = \int_0^{\pi} 2a^2\lambda_0 E_0 d\phi\left[\sin\phi\cos\phi - \sin^2\phi\right]\hat{k}$

$= 2a^2\lambda_0 E_0\left[\int_0^{\pi}\sin\phi\cos\phi d\phi - \int_0^{\pi}\sin^2\phi d\phi\right]\hat{k}$

$= 2a^2\lambda_0 E_0\left[\int_0^{\pi}\frac{\sin 2\phi}{2}d\phi - \int_0^{\pi}\left(\frac{1-\cos 2\phi}{2}\right)d\phi\right]\hat{k}$

$= 2a^2\lambda_0 E_0\left[0 - \frac{\pi}{2}\right]\hat{k} = a^2\lambda_0\pi E_0\,\hat{k}$

18. **(1)**

Electrostatics force on $q = \frac{\lambda q}{2\pi\varepsilon_0 r}$ away from line charge Magnetic force $= \frac{\mu_0\lambda v}{2\pi r}\times q\times v$ away from line charge

$\therefore$ total force $= \frac{\lambda q}{2\pi r}\left[\frac{1}{\varepsilon_0} + \mu_0 v^2\right]$

19. **(4)**

This silvered concavo-convex lens behaves like mirror whose focal length can be calculated by the formula

$\frac{1}{f} = \frac{2}{f_1} + \frac{1}{f_2}$

f_1 = focal length of concave surface.

f_2 = focal length of concave mirror.

$\therefore \frac{1}{f} = \frac{2}{-60} + \frac{1}{-10} = \frac{4}{30}$

$\therefore$ f = –7.5 cm

Using mirror formula

$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{-7.5} = \frac{1}{-x} + \frac{1}{-x} \Rightarrow x = 15$ cm

20. **(3)**

$\lambda_{rod} = \frac{M}{4R}$

now, $dF = \frac{GM\frac{M}{4R}dx}{x^2 + 4R^2}$

Force acting on the rod due to sphere is

$$F = \int_{-2R}^{2R} dF\cos(\theta)$$

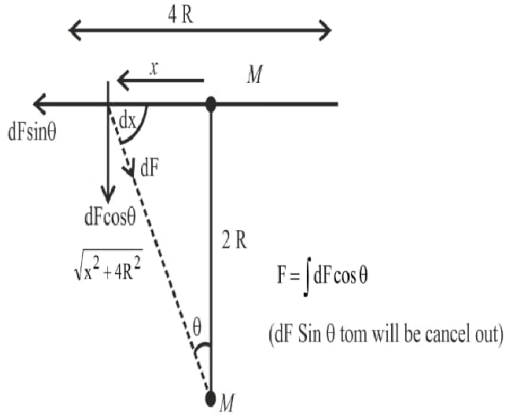
$$= \int_{-2R}^{2R} \frac{GM}{4R}\frac{M}{x^2+4R^2}dx\left(\frac{2R}{\sqrt{x^2+4R^2}}\right)$$

$$= \int_{-2R}^{2R} \frac{GM^2}{2\left(x^2+4R^2\right)^{\frac{3}{2}}}dx$$

after solving this integration we get

$$F = \frac{GM^2}{4\sqrt{2}R^2}\ \text{N}$$

Direction of force will be along $(-\hat{j})$.



21. (494)

Energy of incident photons

$= 13.6\ Z^2$

$= 13.6 \times 2^2$

$= 54.4$ eV

Max K.E. of photoelectrons

$= 54.4 - 5$

$= 49.4$ eV

Stopping potential = 49.4 volt

22. (12)

$F_B = i\ dl \times B$

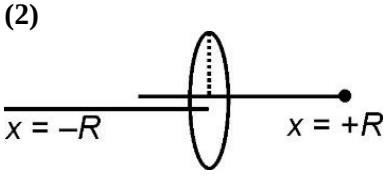
$$F_{\text{net}} = \frac{\mu_0 I_0 \times 2}{2\pi a} Ia - \frac{\mu_0 I_0 \times 2}{2\pi 3a} Ia$$

$$\therefore \quad F_{\text{net}} = \frac{\mu_0 I_0}{2\pi a} 2Ia\,(1-1/3)$$

$$\Rightarrow F_{\text{net}} = \frac{\mu_0 I_0 I}{\pi}\times\frac{2}{3} = 4\times10^{-7}\times\frac{2}{3}\times5\times\frac{9}{10}$$

$= 1.2\ \mu\text{N}$

23. (2)



Alternately, we shall calculate the magnetic field by straight wire with current I and then find its line integral on circle.

$$\therefore \quad \int B\cdot dl = \frac{\mu_0 I}{4\pi R}\cdot\frac{2.1}{\sqrt{2}}\cdot 2\pi R$$

$$\Rightarrow \quad \int B.dl = \frac{\mu_0 I}{\sqrt{2}}$$

24. (6)

$$du = \frac{Y}{2}\epsilon^2\ Adx$$

$$U_0 = \int_0^L \frac{Y}{2}\left(\frac{F(l-x)}{lAY}\right)^2 Adx$$

$$U_0 = \frac{F^2 L}{6AY}$$

25. (12)

$(\Delta x)_{\max} = 2ae = 10\lambda \times 0.3 = 3\lambda$.

For maximum, path difference = $n\lambda$

In a quadrant path difference varies continuously from 3λ to $0 \Rightarrow 4$.

SECTION-II (CHEMISTRY)

26. (2)

I. $n + \ell = 4 + 1 = 5$

II. $n + \ell = 4 + 0 = 4$

III. $n + \ell = 3 + 2 = 5$

IV. $n + \ell = 3 + 1 = 4$

I is having greater n value as compared to III.

II is having greater n value as compared to IV.

27. (1)

$Ba(OH)_2\ (s) \rightleftharpoons Ba^{2+}\ (aq) + 2OH^-\ (aq)$

pH = 12 $\quad\quad \therefore$ pOH = 2

i.e. $[OH^-] = 10^{-2}$

$$\left[Ba^{2+}\right] = \frac{10^{-2}}{2} = 5\times10^{-3}$$

$$K_{sp} = \left[Ba^{2+}\right]\left[OH^-\right]^2 = 5\times10^{-3}\times(10^{-2})^2$$

$= 5 \times 10^{-7}$

28. (3)

Work done = $-P\Delta V = -1(2.5 - 2) = -1 \times 0.5$

$= -0.5$ litre atm

$= -50.63$ J

$Q = \Delta U - W$

$300 = \Delta U - (-50.63)$

$\Delta U = 249.37$ J

29. (3)

The compound must be cyclic and contains a – OH and – CHO group. It must contain six C atoms.

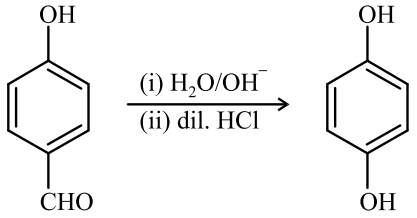
30. (2)

$$\underbrace{BrO_3^- \xrightarrow[n_1=4]{E_1^\circ} BrO^- \xrightarrow[n_2=1]{E_2^\circ} \frac{1}{2}Br_2}_{E^\circ}$$

$$E^\circ = \frac{n_1E_1^\circ + n_2E_2^\circ}{n_1+n_2} = \frac{4\times0.54+1\times0.41}{4+1}$$

= 0.52 V

31. (3)



P does not react with aq. $NaHCO_3$ as it is a weak acid.

32. (1)

Due to +R effect of $-OCH_3$ group, it reacts with electrophile dichlorocarbene in fastest rate.

33. (2)

Atomic radius of Ga is slightly lower than that of Al due to d-Contraction.

34. (2)

N_2O = +1, NO = +2, N_2O_3 = + 3, NO_2 = +4

35. (3)

Guanidine is the most basic compound among the given compounds. Its conjugate acid is most stable due to resonance.

36. (4)

A & B → enantiomers

A & C → diastereomers

37. (3)

BrF_5, ClF_5, IF_5

38. (3)

Because Cr(+VI) has d° configuration hence d-d transition will not possible.

39. (4)

Due to synergic bonding carbon oxygen bond strength decreases.

40. (2)

Test of Phenol, aniline, carboxylic acid and aldehydes.

41. (2)

$$SO_2Cl_2(g) \rightarrow SO_2 + Cl_2(g)$$

0.5	–	–
0.5 – x	x	x

$P_T = 0.5 + x = 0.6$

x = 0.1

$$k = \frac{2.303}{100}\log\frac{0.5}{0.4} = \frac{2.303}{1000}$$

When $P_T = 0.65$

x = 0.15

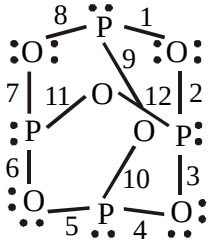
$$\text{Rate} = k\left[P_{SO_2Cl_2}\right] = \frac{2.303}{1000}\times0.35$$

$= 8.06 \times 10^{-4}$ atm sec^{-1}

42. (4)

(1) PH_5 does not exist as there is large difference in energies of s, p and d orbitals, H do not polarize P for intermixing hence it does not undergo Sp^3d hybridisation. In SCl_6, smaller S cannot accomodate six larger Cl^- ions. Fluorine can not expand its octet because it does not have vacant d-orbitals.

(2) P_4O_6



(3) $2PCl_5(s) \rightarrow [PCl_4]^+ + [PCl_6]^-$

43. (3)

(a) Maltose is reducing sugar and gives silver mirror with tollen's reagent.

(b) Sucrose, starch and cellulose are non-reducing sugar.

(c) Starch and cellulose are biopolymers.

(d) Sucrose has α,β (1,2)- glycosidic linkage.

(e) Starch and maltose have α-(1,4)-glycosidic linkage.

(f) Cellulose has β-(1,4)-glycosidic linkage.

44. (3)

$$Ag_2CO_3(s) \xrightarrow{\Delta} 2Ag(s) + CO_2(g) + \frac{1}{2}O_2(g)$$

Weight of pure $Ag_2CO_3 = 3\times\frac{92}{100} = 2.76$ gram

Mole of $Ag_2CO_3 = \frac{2.76}{276} = 0.01$ mol

Mole of evolved CO_2 = mole of Ag_2CO_3

Mole of evolved $O_2 = \frac{1}{2}$ mole of Ag_2CO_3

Total mole of gaseous products $= 0.01 + \frac{1}{2} \times 0.01$

= 001 + 0.005

= 0.015 mol

Volume at STP = 0.015× 22400 = 336 mL

45. (3)

$$\Delta T = \frac{1000 K_b w_1}{m_1 w_2}$$

$$\frac{\Delta T}{K_b}(AB_2) = \frac{1000 \times 6}{m_1 \times 100} = 1, \quad \frac{\Delta T}{K_b}(A_2B) = \frac{1000 \times 9}{m_2 \times 100}$$

$\therefore$ 60 = A + 2B

$\therefore$ 90 = 2A + B

$\therefore$ A = 40, B = 10

46. (2)

If P = initial partial pressure of NO_2

$$2NO_2(g) \rightleftharpoons 2NO(g) + O_2(g)$$

$t = 0$	P		
t_{eqbm}	$P - 2x$	$2x$	x

x = 0.25 $\quad\quad$ $\therefore$ 2x = 0.5

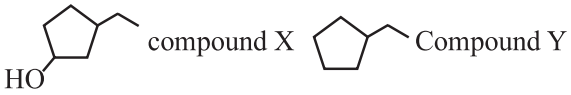
$$K_P \frac{(2x)^2(x)}{(P-2x)^2} \Rightarrow 156.25 = \frac{(0.5)^2(0.25)}{P_{NO_2}^2}$$

$\therefore$ $P_{NO_2} = 0.02$ atm

47. (9)

group = 3, Period = 6 } Lanthanoid series

48. (4)



Therefore molecular weight difference equal to 16 gram/mol.

49. (6)

Only NH_3, H_2O, HF, HCOOH, $B(OH)_3$, CH_3COOH

50. (4)

1º and 2º amines can be distinguished using Azo dye test, Hofmann's mustard oil reaction, Carbyl amine test and Hinsberg reagent.

SECTION-III (MATHEMATICS)

51. (2)

$$\sqrt{x^2-4x+3} + \sqrt{x^2-9} = \sqrt{4x^2-14x+6} \quad \text{....(i)}$$

Here

$x^2 - 4x + 3 \geq 0 \Rightarrow x \in (-\infty, +1] \cup [3, \infty)$

$x^2 - 9 \geq 0 \Rightarrow x \in (-\infty, -3] \cup [3, \infty)$

$4x^2 - 14x + 6 \geq 0$

$\Rightarrow \quad x \in \left(-\infty, +\frac{1}{2}\right] \cup [3, \infty)$

Hence Domain is $x \in (-\infty, -3] \cup [3, \infty)$

Now equation (i) can be written as

$$\sqrt{(x-3)(x-1)} + \sqrt{(x+3)(x-3)} - \sqrt{(x-3)(4x-2)} = 0$$

$$\Rightarrow \quad \sqrt{(x-3)}\left(\sqrt{x-1} + \sqrt{x+3} - \sqrt{4x-2}\right) = 0$$

$\sqrt{x-3} = 0$

$x = 3$

and $\sqrt{x-1} + \sqrt{x+3} - \sqrt{4x-2} = 0$

$\Rightarrow \quad \sqrt{x-1} + \sqrt{x+3} = \sqrt{4x-2}$

$\Rightarrow \quad x - 1 + x + 3 + 2\sqrt{(x-1)(x+3)} = 4x - 2$

$\Rightarrow \quad 4(x-1)(x+3) = (4x - 2 - 2x - 2)^2$

$\Rightarrow \quad 4[x^2 + 3x - x - 3] = (2x - 4)^2$

$\Rightarrow \quad 4[x^2 + 2x - 3] = 4x^2 - 16x + 16$

$\Rightarrow \quad 4[x^2 + 2x - 3] = 4[x^2 - 4x + 4]$

$\Rightarrow \quad x^2 + 2x - 3 = x^2 - 4x + 4$

$\Rightarrow \quad 6x = 7 \Rightarrow x = \frac{7}{6} \notin$ domain

Hence only one solution

52. (3)

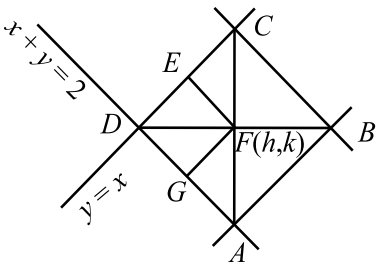
Rewriting the given function, we get,

$f(x) = (x - 6a)^2 + 15 - 2a$

$\because$ $f(x)$ is surjective on R

$\Rightarrow \quad 15 - 2a = 2 \Rightarrow 2a = 13 \Rightarrow a = \frac{13}{2}$

53. (4)



Let diagonals *AC* & *BD* intersect at *F* (*h*, *k*)

$\Rightarrow$ Area of quad *DEFG* is $\frac{1}{4}(20) = 5$

Now distance of (*h*, *k*) from $x + y = 2$ and $x - y = 0$ is $\left|\frac{h+k-2}{\sqrt{2}}\right|$ & $\left|\frac{h-k}{\sqrt{2}}\right|$ respectively.

$\Rightarrow \left|\frac{h+k-2}{\sqrt{2}}\right| \left|\frac{h-k}{\sqrt{2}}\right| = 5$

$\Rightarrow |(h-1)+(k-1)|\,|(h-1)-(k-1)| = 10$

$\Rightarrow |(h-1)^2-(k-1)^2| = 10$

$\Rightarrow (x-1)^2-(y-1)^2 = 10$ or $(x-1)^2-(y-1)^2 = -10$

$\Rightarrow$ Both are hyperbola with eccentricity = $\sqrt{2}$

54. (2)

Given, AB = I

$$AB = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}\begin{bmatrix} -\alpha & 14\alpha & 7\alpha \\ 0 & 1 & 0 \\ \alpha & -4\alpha & -2\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 5\alpha & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10\alpha-2 & 5\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\therefore$ We get, $5\alpha = 1$

$\Rightarrow \alpha = \frac{1}{5}$

Now, $tr(B) = (-\alpha) + 1 + (-2\alpha)$

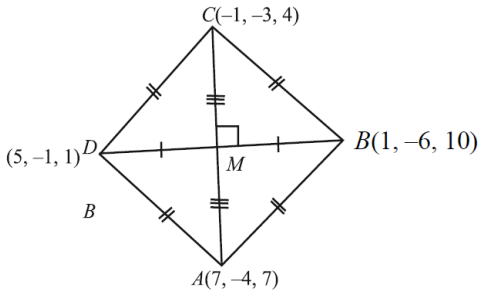
$= 1 - 3\alpha = 1 - \frac{3}{5} = \frac{2}{5}$

55. (2)

Here, $AB = BC = CD = DA (=7)$

Also, $\overline{AB} \parallel \overline{DC}$ and $\overline{BC} \parallel \overline{AD} \neq 0$

As $\overline{AB}.\overline{AD} \neq 0$ and $AC \neq BD$



$\Rightarrow ABCD$ is not square

But $\overline{AC} \perp \overline{BD}$

Hence, $ABCD$ is a rhombus.

56. (1)

P be a point on $(x-1)^2 + (y-1)^2 = 1$

so $P(1 + \cos\theta, 1 + \sin\theta)$

$A(1, 4)$ $B(1, -5)$

$(PA)^2 + (PB)^2$

$= (\cos\theta)^2 + (\sin\theta - 3)^2 + (\cos\theta)^2 + (\sin\theta + 6)^2$

$= 47 + 6\sin\theta$

is maximum if $\sin\theta = 1$

$\Rightarrow$ $\sin\theta = 1$, $\cos\theta = 0$

$P(1, 2)$ $A(1, 4)$ $B(1, -5)$

P, A, B are collinear points.

57. (4)

The given function

$f(x) = (3x^2 + ax - 2 - a)e^x$

$f'(x) = (6x + a)e^x + (3x^2 + ax - 2 - a)e^x$

$f'(x) = [3x^2 + (a+6)x - 2]e^x$

$x = 1$ is critical point

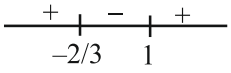
$f'(1) = 0$

$\Rightarrow (3 + a + 6 - 2).e = 0$

$\Rightarrow a = -7$

$\therefore f'(x) = (3x^2 - x - 2)e^x$

$= (3x + 2)(x - 1)e^x$



$x = -\frac{2}{3}$ is point of local maxima

and $x = 1$ is point of local minima

58. (2)

$\cos5\theta = \cos(4\theta + \theta)$

$= \cos4\theta\cos\theta - \sin4\theta\sin\theta$

$= (2[\cos^2\theta - 1]^2 - 1)\cos\theta - 2.2\sin\theta\cos\theta[2\cos^2\theta - 1]\sin\theta$

$= \cos\theta[2(4\cos^4\theta + 1 - 4\cos^2\theta) - 1] - 4\sin^2\theta\cos\theta[2\cos^2\theta - 1]$

$= 8\cos^5\theta - 8\cos^3\theta + 2\cos\theta - \cos\theta - 4(1 - \cos^2\theta)\cos\theta(2\cos^2\theta - 1)$

$= 8\cos^5\theta - 8\cos^3\theta + \cos\theta - 4\cos\theta[2\cos^2\theta - 1 - 2\cos^4\theta - 1 - 2\cos^4\theta + \cos^2\theta]$

$= 16\cos^5\theta - 20\cos^3\theta + 5\cos\theta$

Clearly, $a = 16$, $b = 0$

59. (3)

For $x = n$, $n \in Z$

$\text{LHL} = \lim\limits_{x \to n^-} f(x) = \lim\limits_{x \to n^-} [x-1]\cos\left(\frac{2x-1}{2}\right)\pi = 0$

$\text{RHL} = \lim\limits_{x \to n^+} f(x) = \lim\limits_{x \to n^+} [x-1]\cos\left(\frac{2x-1}{2}\right)\pi = 0$

$f(n) = 0$

$\Rightarrow$ LHL = RHL = $f(n)$

$\Rightarrow$ $f(x)$ is continuous for every real x.

60. (3)

$f(x) = (x-a_1)(x-a_2)(x-a_3) + (a_1 - x) + (a_2 - x) + (a_3 - x)$

Now $f(x) \to -\infty$ as $x \to -\infty$ and $f(x) \to \infty$ are $x \to \infty$

Again $f(a_1) = (a_2 - a_1) + (a_3 - a_1) > 0$

$[\because a_1 < a_2 < a_3]$

$\Rightarrow$ One root belongs to $(-\infty, a_1)$

Also, $f(a_3) = (a_1 - a_3) + (a_2 - a_3) < 0$

$\Rightarrow$ One root belongs to (a_1, a_3) So f(x) = 0 has three distinct real roots

61. (2)

$$\text{Short test distance} = \frac{\begin{vmatrix} 0 & 4 & 1 \\ 3 & -4 & 0 \\ 2\lambda & 3 & -12 \end{vmatrix}}{\begin{vmatrix} \hat{i} & j & k \\ 0 & 4 & 1 \\ 3 & -4 & 0 \end{vmatrix}}$$

$$13 = \frac{|153 + 8\lambda|}{|4\hat{i} + 3j - 12k|}$$

$$= \frac{|153 + 8\lambda|}{13}$$

$$|153 + 8\lambda| = 169$$

$$153 + 8\lambda = 169, -169$$

$$\lambda = \frac{16}{8}, \frac{-322}{8}$$

$$8\left|\sum_{\lambda \in S} \lambda\right| = 306$$

62. (3)

$$\left|(\sin a + \cos a) - \sqrt{2}\right| + \left|b + \frac{1}{b} - 2\right| + |c| \le 0,$$

Each term is zero independently for solutions to exist

$$\sin a + \cos a = \sqrt{2}, b + \frac{1}{b} = 2, c = 0$$

$$\Rightarrow \quad a = \frac{\pi}{4}, b = 1, c = 0$$

$\therefore$ Required determinant =

$$\begin{vmatrix} \sin(4a) & \cos^2 a & 1 \\ b+2 & b^3 & \frac{1}{b} \\ \ln(c+1) & c+5 & c^3+2 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & \frac{1}{2} & 1 \\ 3 & 1 & 1 \\ 0 & 5 & 2 \end{vmatrix} = 12$$

63. (4)

$$\text{General term} = \frac{10!}{r!s!(10-r-s)!} 2^{\frac{r}{2}} 3^{\frac{s}{3}} 5^{\frac{10-r-s}{6}}$$

$(r, s) = (4, 0), (10, 0), (4, 6)$ for rational term.

64. (2)

$$g(x) = \int_x^{\pi/4} \left(f'(t)\sec t + \tan t \sec t f(t) \right) dt$$

$$g(x) = \int_x^{\pi/4} d(f(t) \cdot \sec t) = f(t)\sec t \Big|_x^{\pi/4}$$

$$g(x) = f\left(\frac{\pi}{4}\right)\sec\frac{\pi}{4} - f(x) \cdot \sec x$$

$$g(x) = 2 - f(x)\sec x = 2 - \left(\frac{f(x)}{\cos x}\right)$$

$$\lim_{x \to \frac{\pi}{2}^-} g(x) = 2 - \lim_{x \to \frac{\pi}{2}^-} \left(\frac{f(x)}{\cos x}\right)$$

Using Hospital Rule

$$= 2 - \lim_{x \to \left(\frac{\pi}{2}\right)^-} \frac{f'(x)}{(-\sin x)}$$

$$= 2 + \frac{f'\left(\frac{\pi}{2}\right)}{\sin\frac{\pi}{2}} = 2 + \frac{1}{1} = 3$$

65. (2)

$2x^y + 3y^x = 20$

$$2x^y\left(y \cdot \frac{1}{x} + \ln x \frac{dy}{dx}\right) + 3y^x\left(x\frac{1}{y} \cdot \frac{dy}{dx} + \ln y \cdot 1\right) = 0$$

Put (2, 2)

$$2.4\left(1 + \ln 2 \frac{dy}{dx}\right) + 3.4\left(1.\frac{dy}{dx} + \ln 2\right) = 0$$

$$\frac{dy}{dx}[8\ln 2 + 12] + 8 + 12\ln 2 = 0$$

$$\frac{dy}{dx} = -\left[\frac{2 + 3\ln 2}{3 + 2\ln 2}\right] = -\left[\frac{2 + \ln 8}{3 + \ln 4}\right]$$

66. (1)

Let $I = \int \frac{(1-\cos\theta)^{2/7}}{(1+\cos\theta)^{9/7}} d\theta$

$$I = \int \frac{\left(2\sin^2\theta/2\right)^{2/7}}{\left(2\cos^2\theta/2\right)^{9/2}} d\theta = \frac{1}{2}\int \frac{(\sin\theta/2)^{4/7}}{(\cos\theta/2)^{18/7}} d\theta$$

Put $\frac{\theta}{2} = t \therefore \frac{d\theta}{2} = dt$

$$\Rightarrow I = \int \frac{(\sin t)^{4/7}}{(\cos t)^{18/7}} dt \quad (\text{Here } m + n = -2)$$

$$= \int (\tan t)^{4/7} \sec^2 t dt$$

Put $\tan t = u \therefore \sec^2 t\, dt = du$

$$\Rightarrow I=\int u^{4/7}du=\frac{u^{11/7}}{11/7}+c=\frac{7}{11}\left(\tan t\right)^{11/7}+C$$

$$=\frac{7}{11}\left(\tan\frac{\theta}{2}\right)^{11/7}+C$$

67. **(1)**

P (odd number 7 times) = P (odd number 9 times)

$${}^{n}C_7\left(\frac{1}{2}\right)^7\left(\frac{1}{2}\right)^{n-7}={}^{n}C_9\left(\frac{1}{2}\right)^9\left(\frac{1}{2}\right)^{n-9}$$

${}^{n}C_7={}^{n}C_9$

$\Rightarrow \quad n=16$

Required

$$P={}^{16}C_2\times\left(\frac{1}{2}\right)^{16}$$

$$=\frac{16\cdot15}{2}\times\frac{1}{2^{16}}=\frac{15}{2^{13}}$$

$$\Rightarrow \quad \frac{60}{2^{15}}\Rightarrow k=60$$

68. **(3)**

$$\frac{(1-x)(1-y)(1-z)}{xyz}$$

$$=\frac{(y+z)(z+x)(x+y)}{xyz}\geq\frac{\left(2\sqrt{yz}\right)\left(2\sqrt{zx}\right)\left(2\sqrt{xy}\right)}{xyz}$$

$= 8$

69. **(2)**

$$\left(1-x^2\right)\frac{dy}{dx}=xy+\left(x^3+2\right)\sqrt{1-x^2}$$

$$\Rightarrow\frac{dy}{dx}+\left(\frac{-x}{1-x^2}\right)y=\frac{x^3+2}{\sqrt{1-x^2}}$$

If $=e^{\int\frac{-x}{1-x^2}dx}=\sqrt{1-x^2}$

$$y(x)\cdot\sqrt{1-x^2}=\frac{x^4}{4}+2x+c$$

$$y(0)=0\Rightarrow c=0$$

$$\sqrt{1-x^2}\,y(x)=\frac{x^4}{4}+2x$$

required value $=\int_{-1/2}^{1/2}\left(\frac{x^4}{4}+2x\right)dx-\frac{1}{4}\cdot2\int_0^{1/2}x^4dx$

$$=\frac{1}{10}\left(x^5\right)_0^{1/2}=\frac{1}{320}$$

$$k^{-1}=320$$

70. **(3)**

$\log_{10} x = A \qquad x > 0$

$\log_{10}(x-2) = B,\ x-2>0 \Rightarrow x>2$

$\Rightarrow \quad A^2-3AB+2B^2<0$

$\Rightarrow \quad (A-2B)\,(A-B)<0$

$\Rightarrow (\log x-2\log(x-2))\,(\log x-\log(x-2))<0$

Case-I: $\log x-2\log(x-2)<2$

and $\log x-\log(x-2)>0$

$\Rightarrow \quad x>2$ and $(x-2)^2>x$

$\Rightarrow \quad x^2-5x+4>0$

$\Rightarrow \quad x>4$

and

$\log x>\log(x-2)$

$\Rightarrow \quad x\in R$

$\therefore \quad x>4$

Case-II: $\log x-2\log(x-2)>0$

and $\log x-\log(x-2)<0$

$\Rightarrow \quad x\in\phi$

From both the cases we get $x\in(4,\infty)$

71. **(9)**

Each element of set A can be distributed in two sets B and C by 3 ways.

(The three ways are either only in B or only in C or in none of B and C).

So, the number of ways distributing = $(3)^4$

$\Rightarrow$ 81 ways.

72. **(9)**

$z = 1 + i$

$$z_1=\frac{1+1\bar{z}}{\bar{z}(1-z)+\frac{1}{z}}$$

$$z_1=\frac{1+i(1-i)}{(1-i)(1-1-i)+\frac{1}{1+i}}=\frac{1+i-i^2}{(1-i)(-i)+\frac{1-i}{2}}$$

$$\frac{2+i}{\frac{-3i-1}{2}}=\frac{4+2i}{-3i-1}=\frac{-(4+2i)(3i-1)}{(3i)^2-(1)^2}=-1+i$$

$$\text{Arg }(z_1)=\frac{3\pi}{4}$$

$$\therefore\ \frac{12}{\pi}\arg(z_1)=\frac{12}{\pi}\times\frac{3\pi}{4}=9$$

73. **(91)**

$$y=\sum_{k=1}^{6}k\cos^{-1}\left\{\frac{3}{5}\cos kx-\frac{4}{5}\sin kx\right\}$$

Let $\cos a=\frac{3}{5}$ and $\sin a=\frac{4}{5}$

$$\therefore\ y=\sum_{k=1}^{6}k\cos^{-1}\{\cos a\cos kx-\sin a\sin kx\}$$

$$=\sum_{k=1}^{6}k\cos^{-1}(\cos(kx+a))$$

$$= \sum_{k=1}^{6} k\,(kx + a) = \sum_{k=1}^{6} (k^2 x + ak)$$

$$\therefore \frac{dy}{dx} = \sum_{k=1}^{6} k^2 = \frac{6(7)(13)}{6} = 91$$

74. (6)

$$\frac{S_n}{\Delta_n} = \frac{2n+5}{3n+2}$$

$$\frac{\frac{n}{2}\left[2A + (n-1)D\right]}{\frac{n}{2}\left[2a + (n-1)d\right]} = \frac{2n+5}{3n+2}$$

$$\frac{A + \left(\frac{n-1}{2}\right)D}{a + \left(\frac{n-1}{2}\right)d} = \frac{2n+5}{3n+2}$$

$$\frac{n-1}{2} \to n-1 \Rightarrow n \to 2n-1$$

$$\therefore \quad \frac{T_n}{t_n} = \frac{A + (n-1)D}{a + (n-1)d} = \frac{2(2n-1)+5}{3(2n-1)+2}$$

$$= \frac{4n+3}{6n-1}$$

Let $\to T_n = k\,(4n + 3)$

$t_n = k\,(6n - 1)$

Now comparing gives

For $T_n \Rightarrow D = 4k$, $A - D = 3k$

$A = 7k$

For $t_n \Rightarrow d = 6k$, $a - d = -k$

$a = 5k$

$$\therefore \quad \frac{T_{11}}{t_{10}} = \frac{7k + 10 \cdot 4k}{5k + 9 \cdot 6k} = \frac{47}{59} = \frac{m_1}{m_2}$$

$$\therefore \quad \frac{m_2 - m_1}{2} = \frac{59 - 47}{2} = \frac{12}{2} = 6$$

75. (2)

$$\lim_{x \to \infty} \frac{x^2 f(x)}{2x^5 + 3} = 6$$

$\Rightarrow f(x)$ is a cubic polynominal with leading Coefficent 12

Let $f(x) - (2x+1) = 12(x-1)(x-3)(x-5)$

$f(x) = 12(x-1)(x-3)(x-5) + (2x+1)$

Now $\frac{f(6) + 5f(4)}{29} = \frac{58}{29} = 2$



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if